

EnterprateAI Shared Architecture and Feature Integration Blueprint

Version 0.1 | 30 September 2026 | Proposed design for product and engineering review

1 Purpose and status

Establish the common foundation for Funding Readiness Check, Launch Readiness Check, Adaptive Dashboard and Agentic Orchestration. Deliver the features through four linked PRDs, with this document owning shared architectural requirements.

This blueprint is grounded in the supplied Updated Business Ontology and five engine specifications. It does not establish what is deployed. Database schemas, APIs, source code, authentication, entitlements and production behaviour must be verified before implementation estimates and migrations are finalised. Proposed entity and endpoint names below are conceptual contracts to map onto the current implementation.

Product direction: users supply business information once, receive explainable assessments, see relevant priorities and can delegate supported actions under explicit permissions. Retain EnterprateAI's decision intelligence focus and reuse its existing business operations.

2 Architecture decisions

ID	Proposed decision	Practical consequence
AD-01	Reuse the current application stack and system of record	Avoid a framework or database migration solely for these features
AD-02	Extend the existing six ontology domains	Add related entities where necessary without creating isolated feature databases
AD-03	Share a versioned assessment service	Funding and launch checks use common evidence, results, risks and recommendations
AD-04	Preserve deterministic calculation ownership	Engines calculate; language models explain, extract proposed inputs and plan permitted actions
AD-05	Separate stage, account scope, role and subscription	Startup, Small Business and Multi-Entity pathways remain flexible
AD-06	Separate recommendations from executable actions	A recommendation can exist without granting an agent execution authority
AD-07	Use a common agent runtime and tool registry	New agents reuse permissions, approvals, execution state and auditing
AD-08	Preserve assessment snapshots and ruleset versions	Historical decisions remain reproducible after data or rules change

ID	Proposed decision	Practical consequence
AD-09	Build modular boundaries inside the existing deployment first	Extract independently scaled workers/services when operational evidence warrants it

3 Logical architecture and ownership

Layer	Owns	Interfaces with
Experience	Onboarding, three dashboard pathways, assessment journeys, action and approval views	Application APIs
Application services	Funding cases, launch initiatives, dashboard composition, business operations	Shared data, assessment and workflow services
Intelligence	Input validation, computation library, engine orchestration, readiness rules, interpretation	Versioned business snapshots and assessment records
Agent orchestration	Goal decomposition, workflow state, tool selection, approval checkpoints, execution recovery	Registered tools and authorised application APIs
Shared platform	Identity, permissions, evidence, events, background jobs, audit, usage and observability	All modules
Data	Existing ontology and business records, assessment history, scenarios and execution history	Controlled repositories/services

Engine orchestration determines which calculations are required and persists their outputs. Agent orchestration coordinates work towards a user goal. Agents call engine tools when needed; they cannot override calculated scores or bypass application permissions.

Example: changed expense data makes a cash assessment stale. A background job recalculates affected outputs. The dashboard shows the resulting risk and recommended action. A user may then delegate preparation of a cost-reduction scenario. Running that scenario does not change live expenses.

4 Reuse and extension map

Existing documented structure	Intended reuse	Extension to investigate
Business, BusinessProfile	Identity, stage, country, sector and currency	Account-to-business membership, scoped permissions, stage history
Offering, PricingModel, DemandEvidence	Launch economics and market evidence	Link a launch initiative to one or more offerings; broaden evidence references

Existing documented structure	Intended reuse	Extension to investigate
FinancialPeriod, CashAccount, RevenueRecord, ExpenseRecord, CashflowSnapshot	Funding and launch financial inputs	Explicit actual/forecast/estimate basis, snapshot lineage and data freshness
Staff, Role, Process, Dependency, CapacityProfile	Delivery and operational readiness	Launch prerequisites and evidence of completion
Assessment, AssessmentMetric, AssessmentScore	Shared assessment persistence	Subject type/ID, input snapshot reference, criteria version and applicability
RiskFlag, Recommendation, Interpretation	Findings, explanations and next actions	Recommendation-to-action link, deduplication and outcome tracking
Scenario, ScenarioChange, SimulationRun, SimulationResult	Funding and launch scenarios	Immutable baseline reference and approved adoption workflow

Proposed additions, subject to the production inventory:

- **FundingCase and UseOfFundsItem:** funding route, target amount/currency, timing, purpose, milestones, forecast references and allocations. A business may have multiple cases over time.
- **LaunchInitiative and LaunchMilestone:** initial business, new offering or market launch; target date, owner, scope and prerequisite dependencies.
- **EvidenceItem and EvidenceLink:** source, claim supported, attachment/reference, effective date, verification status and freshness. Extend existing evidence infrastructure if available. Preserve DemandEvidence for its established commercial meaning.
- **AssessmentDefinition and CriterionResult:** versioned criteria, applicability, weight where justified, blocker policy, result and evidence links. Configuration must be schema-validated, not arbitrary executable code.
- **ActionItem:** owner, due date, source recommendation, status and completion evidence. This is a focused follow-through record, not a general project management module.
- **AgentDefinition, ToolDefinition, WorkflowDefinition, WorkflowRun, StepRun and Approval:** reusable execution infrastructure with versioned contracts.
- **DashboardPreference and WidgetDefinition:** stable widget identifiers, eligibility rules and user choices. Compute dashboard views from authoritative results rather than creating another score store.

5 Shared assessment contract

Every persisted result must identify the business, subject, assessment type, input snapshot, ruleset version, engine versions, data timestamp and creation timestamp. Record individual criterion results, supporting evidence, missing inputs, blockers, metrics, risk flags and recommendation references.

Keep separate concepts:

Concept	Proposed values or treatment
Execution status	queued, running, succeeded, failed, cancelled

Concept	Proposed values or treatment
Criterion state	met, partially_met, unmet, unknown, not_applicable
Readiness classification	ready, conditionally_ready, not_ready, insufficient_evidence
Data basis	actual, forecast, estimate; mixed results retain field-level basis
Freshness	current or stale, with reason and timestamp
Confidence	Separate from readiness; derived from evidence quality, completeness and age

Weights and thresholds belong in each PRD and a reviewed ruleset. Do not invent a production scoring model by averaging existing engine scores. A critical unmet prerequisite blocks a ready classification. Unknown mandatory evidence produces insufficient evidence unless a confirmed blocker already establishes not-ready. Non-applicable criteria require an applicability reason and must not inflate results through arbitrary exclusion.

An LLM can suggest extracted data, interpretations and actions. Material extracted inputs require validation and provenance before scoring. If narrative generation fails, the deterministic assessment must remain usable.

6 Funding Readiness scope

Question: how prepared is this business to pursue a selected funding route, and what is missing?

Reuse financial, viability, survival, stability and growth outputs only where applicable to stage and route. Add criteria for funding purpose and amount, use of funds, evidence of demand/traction, team capability, forecasts, milestones and document completeness. Equity, debt and grant routes require separate criterion profiles. A route that has not been implemented must be clearly unavailable.

First release proposal: equity preparation for UK service and software businesses, with other routes supported by the architecture and delivered after their criteria are defined. Confirm this commercial scope during PRD development. Readiness indicates preparation against the selected criteria; it does not represent a funding approval probability.

The user selects a funding case, confirms reused data, fills relevant gaps, receives an assessment and works through prioritised actions. Documents reference the same case and assessment snapshot. Missing facts become explicit gaps rather than generated claims. Funding scenarios model amount, timing, allocation and operating assumptions without overwriting the live baseline.

The supplied UK Endorsing Services documents are candidate references for a later endorsement-specific requirements profile. Endorsement must not be conflated with funding readiness. Their current requirements must be separately reviewed before any such profile is released.

7 Launch Readiness scope

Question: what must be completed before this business or offering can launch?

Assess demand validation, offer and pricing, economics, cash preparation, delivery capacity, operational prerequisites and route-to-market preparation. Applicable sector/country requirements need maintained criterion profiles and named evidence sources before release.

First release: an initial service/software business launch, using the documented service viability model where applicable. New offerings and market launches share the initiative structure but can follow in later releases. Do not apply service-specific formulas to manufacturing or inventory-heavy businesses without an appropriate model.

The user creates an initiative, sets a target date, reviews prefilled evidence, completes missing inputs and receives blockers and next actions. Completing a task does not automatically establish readiness: evidence must be updated and the relevant assessment rerun. Historical operating metrics unavailable for a startup must remain unavailable or explicitly estimated.

8 Adaptive Dashboard and the three pathways

Model four independent dimensions: business stage, account scope, user role and subscription entitlements. Multi-Entity is an account view that can contain startups and operating businesses. Switching pathway must not recreate or duplicate the business.

Pathway	Default priorities	Supporting views
Startup	Next actions, launch blockers, viability and forecast cash position	Funding preparation, business plan, scenarios and business operations
Small Business	Cash and risk alerts, operational actions, recommendations	Fragility Index, simulations, funding, growth and new launch initiatives
Multi-Entity	Exceptions needing attention, entity comparison and approval queue	Business drill-down, readiness status, scenario and action summaries

Preserve Business Operations in navigation. Preserve access to risk alerts, Fragility Index, adaptive recommendations and scenario simulation where available and entitled. An operating business can use idea validation or launch a new service.

Use an initial maximum of five primary cards as a design proposal, with detail available on demand. Priority order: critical risks/blockers, time-sensitive approvals, chosen goal, then supporting insights. Users can pin cards; explain material changes without continuously rearranging the page. Explicitly show empty, insufficient-data, stale, loading and failed states.

Multi-Entity phase one should compare entity results and show risk counts. Financial totals require consistent periods, exchange-rate provenance and double-counting rules. Accounting-style consolidation and intercompany eliminations are outside this initial scope. Never average readiness or Fragility Index scores by default. Permission checks apply before both detail retrieval and aggregation.

9 Agentic Orchestration contract

Every workflow run has a requesting actor, business scope, goal, definition version, permitted tools, execution budget and correlation ID. Each tool defines its input/output schema, effect type, required permissions, timeout and retry/duplicate-prevention behaviour.

Execution progression: created → running → awaiting approval when required → running → succeeded, failed or cancelled. Persist steps so recovery resumes from the last confirmed state. Recheck authority and relevant

business data before an external action, including after approval. Approvals bind to a specific action payload/version; material edits invalidate them.

Support assistance, draft preparation, execution with approval and bounded automatic execution as distinct permissions. Do not treat conversational interest or a recommendation as permission to send, sign, pay or submit. Credentials remain in an appropriate secrets store. External enquiries and documents are untrusted input and cannot alter tool policies.

First workflow: enquiry classification and quotation preparation. It can identify a quotation, proposal, complaint or other enquiry and route accordingly. Reuse catalogue, customer and quotation functions. Begin with drafts and user-approved sending. Later expand to quotation → contract → invoice or quotation → invoice, with payments and receipts tied to confirmed payment events.

Prevent duplicate sends and duplicate documents with idempotency keys and reconciliation after uncertain provider responses. Failed steps expose a useful status and recovery action. Record actual outcomes, costs and tool calls without logging secrets or unnecessary customer data.

10 APIs, events and expansion

Map these logical operations onto current endpoints: start/read assessment; create/update funding case; create/update launch initiative; retrieve dashboard; start/read/cancel workflow; approve/reject an action. Return asynchronous run IDs for long-running work. Enforce business scope at the service/data boundary as well as the interface.

Suggested events: BusinessDataChanged, EvidenceUpdated, AssessmentCompleted, AssessmentInvalidated, RecommendationCreated, ActionCompleted and WorkflowStatusChanged. Each includes event ID, tenant/business scope, schema version, timestamp and causation/correlation IDs.

Publish committed changes reliably through the current equivalent of a transactional outbox. Consumers must tolerate duplicate delivery. Coalesce bursts of changes, invalidate affected assessments and avoid assessment-trigger loops. Refresh only affected outputs; historical snapshots remain immutable.

Register new criterion profiles, widgets, agent tools and workflows through versioned definitions. Use background workers, indexed scoped queries, pagination and permission-aware caching to support growth. Choose queue technology and deployment topology after reviewing the current stack. Record response time and job latency targets in PRDs after measuring the baseline.

11 Source validation issues to resolve

These are specification findings, not confirmed production defects.

Finding	Required resolution before reuse
Service Viability §2 labels break-even sales divided by expected monthly sales as break-even months	Clarify it as a monthly break-even coverage relationship. Time to profitability/payback needs a time-based forecast and explicit initial investment assumptions
Survival §3 defines net cashflow using revenue minus expenses	Establish actual cash receipts/payments versus accrual values; clearly label estimates

Finding	Required resolution before reuse
Survival §10 sample sub-scores 40, 30, 50 give 40 under §5 weights, while sample reports 41	Replace illustrative fixtures with verified calculations; reconcile sub-scores against scoring bands
Stability §9 sample sub-scores 40, 30, 40, 70 give 45 under §5 weights, while sample reports 43	Reconcile examples, classifications and expected test outputs
Several scoring bands share boundaries; some rules have unspecified penalties or thresholds	Define inclusive/exclusive boundaries, precedence, caps, defaults and complete classification coverage
Zero revenue, zero capacity, missing history and non-burning runway need explicit semantics	Prevent division errors, misleading zeroes and non-JSON infinity values; return typed unavailable/unbounded states with reasons
Growth master score combines engines with fixed weights	Define applicability for pre-revenue businesses and missing engines; do not silently renormalise
Fragility Index is a required product dependency, but its formula is not defined in the reviewed engine set	Obtain its authoritative specification and verify production behaviour before changing or reusing it

12 Delivery sequence and release gates

Phase	Deliverable	Exit condition
0	Production inventory and reuse matrix	Existing, extend and new items identified with code/schema evidence; scoring discrepancies assigned
1	Shared assessment, scope, evidence and execution contracts	Contract examples reviewed; migration and isolation strategy defined; all four PRDs drafted against them
2	Launch check plus Startup dashboard	One complete journey from reused data to blockers, actions and reassessment works
3	Funding check plus Small Business dashboard	Selected funding route and relevant existing workflows pass acceptance checks
4	First agent workflow	Drafting, approval, sending, failure recovery and duplicate protection demonstrated
5	Multi-Entity experience	Scoped comparison, drill-down and aggregation permissions validated

Entity isolation and agent interfaces are foundation work even though their full experiences ship later. Phase order is a recommendation; capacity and user testing determine dates.

Use additive migrations, verified backfills, feature flags and pilot cohorts. Preserve old API contracts while introducing new ones. Roll back application behaviour with flags; avoid destructive migration reversal. Do not overwrite historic assessment results when rules change.

13 Shared acceptance criteria

1. Identical validated inputs and ruleset produce identical numerical results.
2. A startup without historical revenue receives an explicit insufficient-data/forecast treatment, never fabricated operating history.
3. A high aggregate score cannot conceal a failed critical readiness criterion.
4. Updating evidence invalidates the appropriate current result and preserves the prior snapshot.
5. A simulation leaves live business inputs unchanged.
6. An entity-scoped user cannot access another entity through URLs, jobs, exports, search, caches or portfolio totals.
7. Changing business stage preserves records and user access to relevant tools.
8. Duplicate events or retries do not create duplicate quotations or external sends.
9. A modified or expired approval cannot authorise a changed action.
10. LLM or integration failure leaves valid deterministic results and recoverable workflow state available.
11. Existing invoice, quotation, contract and dashboard journeys continue to function during staged rollout.

Track assessment completion, time to first useful result, data reuse, blocker resolution, repeated assessment use, recommendation follow-through, agent task completion, correction rate and cost per successful workflow. Define targets after baseline measurement; signup volume alone does not establish usefulness.

14 Immediate handoff and PRD package

The development lead should provide a schema/migration inventory, API catalogue, authentication and membership model, scoring implementations and tests, dashboard routes/components, current job infrastructure and integration interfaces. Use sanitised examples; production credentials and customer records are unnecessary.

Then prepare four PRDs. Each must contain the user problem, journeys, MVP/exclusions, source-to-field mapping, rule definitions, UX states, API/event dependencies, permission matrix, migration plan, acceptance examples, instrumentation and rollout gates. Link shared rules to this blueprint and maintain a decision log for changes.

Decisions still requiring product/engineering resolution: first funding route; service/software-only launch scope; account roles; existing subscription entitlements; permitted agent autonomy; authoritative Fragility Index formula; and measurable performance budgets. These do not block drafting, but their owners must resolve them before the affected release.

Source register

- Updated Business Ontology.pdf: §§1–3 domains, entities and relationships; §§5–9 API, history, simulation, scoring and architecture.
- Service Idea Viability Engine.pdf: §§1–4 inputs, calculations and scoring.
- SURVIVAL ENGINE.pdf: §§3–6 inputs, calculations and classification; §10 sample output.
- STABILITY ENGINE.pdf: §§3–7 inputs, calculations, classification and flags; §9 sample output.
- GROWTH ENGINE.pdf: inputs, scoring, scenarios and master-score sections.
- REAL-TIME SIMULATION ENGINE.pdf: §§2 and 4–7 calculation reuse, execution flow and result comparison.

Other supplied business-plan and endorsement materials are retained as supporting inputs for subsequent feature-specific work; they are not treated here as verified current external requirements.